A Comprehensive Analysis of CATL: Corporate Strategy and Investor Sentiment in Stock Pricing

Ji Yuzhu[1]

International College Beijing

China Agricultural University, Beijing, 100083, P. R. China

Abstract

This paper presents a two-part analysis of Contemporary Amperex Technology Co., Limited (CATL), the global leader in the new-energy battery industry. Part I evaluates the company's strategic position through PEST, Porter's Five Forces, and SWOT frameworks, complemented by valuation metrics; it finds that CATL's leadership rests on scale, R&D intensity, and supply-chain integration, supported by domestic policy but challenged by international regulation, intense competition, and the risk of technological substitution. Part II empirically examines how investor sentiment relates to CATL's stock returns, comparing institutional research-report sentiment with retail forum (Guba) sentiment, testing supply-chain sentiment mediation, and evaluating daily machine-learning predictions. At the monthly frequency, sentiment shows no significant linear link with returns while competitor returns co-move strongly with CATL; at the daily frequency, sentiment persistence carries weak but detectable predictive information. Together, the two parts connect CATL's fundamental strengths with the market's pricing behaviour.

Key words: CATL; new energy battery; PEST analysis; Porter's Five Forces; SWOT analysis; investor sentiment; Guba; supply-chain spillover; machine learning

Part I Corporate Analysis of CATL: Strategic Positioning

1 Corporate Profile and Business Operations

As a globally leading innovator in new energy technologies, Contemporary Amperex Technology Co., Limited (CATL) has established a highly vertically integrated ecosystem at its business core. According to market data providers, the company achieved main business revenue of RMB 178.9 billion in the first half of 2025.

From a product-structure perspective, the power battery system remains its primary revenue pillar, generating RMB 131.6 billion with a gross margin of 22.41%. The energy storage battery system delivered an impressive performance, with revenue reaching RMB 28.4 billion and a gross margin (25.52%) surpassing that of the power battery business. Additionally, businesses such as battery materials and recycling (RMB 7.887 billion) and battery mineral resources (RMB 3.361 billion) have further reinforced its supply-chain resilience.

In terms of industry and regional distribution, the company's operations are primarily concentrated in the electrical machinery and equipment manufacturing industry (RMB 175.5 billion), while the mining and smelting sector accounts for a relatively small proportion. Regarding market distribution, domestic market revenue stood at RMB 117.7 billion (gross margin: 22.94%), whereas overseas market revenue reached RMB 61.21 billion, with international operations demonstrating stronger profitability at a gross margin of 29.02%.

---

[1] * Corresponding author: Feng Weizhe.

2 PEST Analysis

2.1 Political

Politically, CATL operates amid a dichotomy of robust domestic support and escalating international regulatory hurdles.

Domestically, national macro-strategies and industry standards provide a fortified foundation. The "Dual Carbon" goals [1] serve as the strategic blueprint for long-term growth, while consumer incentives like NEV purchase subsidies [2] directly catalyze market demand. Simultaneously, the Lithium-ion Battery Industry Specification Conditions [3] have raised technical and safety thresholds, enabling industry leaders like CATL to consolidate market dominance through superior scale and technology.

Internationally, the landscape is shaped by concurrent trade barriers and "green" entry requirements. The U.S. Inflation Reduction Act [4] presents significant challenges to global supply-chain layouts through strict localization mandates. Conversely, the EU Battery Regulation [5] imposes rigorous compliance regarding carbon footprints, recycled content, and "battery passports." Although these regulations increase short-term costs, they compel CATL to accelerate innovations in green manufacturing and digital management, ultimately bolstering its long-term global competitiveness.

2.2 Economic

Economically, CATL navigates a landscape defined by structural opportunities and cyclical challenges.

On one hand, industry growth remains highly certain. Driven by projections that China's NEV penetration will surpass 55% by 2025 and 70% by 2030 [6], and bolstered by the Chinese auto industry's achievement of "two consecutive global firsts" [7], this robust structural expansion provides significant long-term headroom that effectively offsets cyclical economic fluctuations.

On the other hand, CATL proactively mitigates two primary challenges. First, to counter the impact of volatile upstream raw material prices (especially lithium) on gross margins, the company employs multi-dimensional strategies: securing resources via mineral investments, establishing recycling systems, and optimizing product structures [8]. Second, to manage interest and exchange rate risks stemming from overseas expansion, CATL utilizes foreign exchange hedging and maintains ample cash reserves to ensure financial resilience [9].

In summary, through forward-looking supply-chain integration and prudent risk management, CATL is successfully converting short-term market volatility into long-term cost leadership and operational stability.

2.3 Social

CATL is capturing structural dividends from evolving consumption patterns, talent-based barriers, and regional market variations.

First, the consumer market is characterized by robust demand and deepening structural shifts. The rise of Gen Z as a core consumer group has shifted the competitive focus from basic battery range to high-tech experiences, such as autonomous driving [10]. Simultaneously, policy targets aiming for 80% electrification in the public sector create significant, predictable growth for commercial batteries [11]. Together, these trends drive the market toward "brand mindshare" and "full-scenario coverage."

Second, elite talent reserves have become the ultimate industry barrier. As a technology-intensive sector, innovation speed depends on "talent density." CATL's R&D team of over 20,000 significantly outpaces international competitors [12]. This massive talent advantage enables simultaneous breakthroughs across multiple technical routes, building an impenetrable patent moat.

Finally, regional climate variations necessitate a diversified technical approach. To address range anxiety in cold northern regions, Plug-in Hybrid (PHEV) and Extended-Range (EREV) models have emerged as high-growth segments [13]. This market reality requires industry leaders to maintain a comprehensive product matrix covering both pure electric and hybrid technologies.

In summary, by mastering consumer trends, sustaining overwhelming talent investment, and securing a versatile technical supply, CATL has successfully converted socio-cultural logic—demand upgrades, talent competition, and market segmentation—into a systemic advantage that solidifies its global leadership.

2.4 Technological

Technologically, CATL aligns with industry evolution through high-intensity R&D investment and proactive multi-pathway layouts.

First, the company's core competitiveness is rooted in a systematic R&D framework. With a 2023 R&D expenditure of RMB 18.356 billion and a team of over 20,000 researchers—of which more than 20% hold master's or doctoral degrees—CATL employs a "capital + talent" saturation strategy [14]. This approach drastically accelerates the transition from lab results to mass production. A definitive example is the Qilin Battery (third-generation CTP); by achieving an energy density of 255 Wh/kg and a 10-minute charge to 80%, it provides a systemic solution to critical market pain points such as range and charging speed [15].

Second, CATL builds a formidable moat through a triad of "active production, strategic reserve, and ongoing research." Moving beyond single-product reliance, the company simultaneously advances the industrialization of Qilin, Condensed, Sodium-ion, and M3P batteries [14]. This diversified matrix ensures comprehensive coverage across all application scenarios, creating a technical barrier that is difficult for competitors to replicate in the short term. Furthermore, CATL maintains a close watch on national-level strategic plans, such as the China Hydrogen Energy Development Report [16], effectively mitigating the risk of technical paradigm shifts.

In summary, CATL leverages massive R&D to drive continuous product iteration while utilizing multi-pathway research to widen its strategic moat. This dual approach ensures current market dominance while preparing the company for future technological transitions.

3 Porter's Five Forces Analysis

3.1 Rivalry Among Existing Competitors (Intensity: High)

While CATL retained its position as the global leader in power battery installations in 2025, the competitive landscape is shifting from "single-player dominance" to a "multi-polar" environment. CATL maintains a global market share of approximately 39% [17]. However, BYD is rapidly narrowing the gap through its vertical integration model and the cost advantages of its Lithium Iron Phosphate (LFP) batteries. Furthermore, second-tier manufacturers such as CALB and EVE Energy are eroding CATL's domestic share through price wars and customized services, exerting downward pressure on industry-wide gross margins [18].

3.2 Bargaining Power of Suppliers (Intensity: Extremely Low)

CATL has significantly weakened the bargaining leverage of upstream suppliers through a deep vertical integration strategy [19]. By securing core mineral resources like nickel globally and accelerating integrated supply-chain projects in regions like Indonesia, CATL has achieved strong penetration into the raw material sector. Additionally, the large-scale deployment of diversified technical routes, such as sodium-ion batteries, has effectively reduced reliance on any single mineral supplier [20]. Leveraging its

massive procurement scale and long-term framework agreements, CATL commands immense pricing power and payment-term control over midstream suppliers of electrolytes and separators, maintaining high cost-resilience against raw material fluctuations [21].

3.3 Bargaining Power of Buyers / Automakers (Intensity: Medium)

Downstream automakers exhibit a paradoxical stance in their negotiations with CATL. On one hand, core clients like Tesla and NIO are pursuing "de-CATL-ization" strategies—diversifying their supply chains or even developing in-house batteries to mitigate supply risks [20]. On the other hand, CATL's absolute technical barriers in ultra-fast charging and high energy density (e.g., the Shenxing and Qilin batteries) ensure that high-end models remain highly dependent on its technology [22]. While this technical premium offsets some buyer power, high price sensitivity in the mid-to-low-end market is forcing CATL to offer competitive pricing policies or proactive price adjustments [23].

3.4 Threat of Substitutes (Intensity: Low to Medium)

While there is currently no disruptive "total replacement" for power batteries, "internal cannibalization" between different technical routes poses a primary threat. Sodium-ion and solid-state batteries are the most critical focus areas. CATL has addressed lithium scarcity with its Naxtra sodium-ion series, planning for larger-scale application in 2026 to consolidate its position in the entry-level market [24]. Regarding solid-state batteries, despite rapid progress claims by competitors, CATL's R&D team of over 1,000 people has moved into the 20 Ah sample pilot production stage, successfully converting "substitution risk" into a technological moat [25].

3.5 Threat of New Entrants (Intensity: Extremely Low)

The power battery industry has entered an oligopolistic stage driven by dual barriers of "Capital + Technology," making the entry threshold for new players exceptionally high.

- **Technical Barriers:** CATL's global patent moat in sodium-ion batteries, fast-charging, and recycling has set international standards, forcing potential competitors to face prohibitive technology-evasion costs [24].
- **Economies of Scale:** Head manufacturers have driven the cost per watt down to levels unattainable for newcomers through high automation and vertical integration. Consequently, startups with low utilization rates face severe loss risks [19].
- **Geographic and Policy Barriers:** Deeply integrated industrial park layouts in Europe (e.g., Hungary) and Southeast Asia have established formidable entry barriers, making it nearly impossible for cross-industry entrants to challenge CATL's market position [20].

4 SWOT Analysis

4.1 Strengths (Internal Positive Factors)

- **Absolute Market Leadership and Economies of Scale:** As of H1 2025, CATL maintains a global market share of 38.1%, consistently ranking first worldwide [26]. This scale allows the company to minimize upstream costs through massive procurement and achieve peak manufacturing efficiency.
- **Multi-dimensional Technological Moat:** CATL possesses a comprehensive R&D system covering "materials, structure, and business models." Beyond the mature Qilin and Shenxing lines,

the Naxtra sodium-ion battery—which passed the 2026 national standards—boasts an energy density of 175 Wh/kg and operates reliably from −40 °C to 70 °C [27].

- **Robust Financial Performance:** In Q3 2025, net profit grew by 41.2%, with net margins climbing to 19.1% [28]. These high returns provide a steady capital base for an annual R&D investment exceeding RMB 10 billion.
- **Superior Circular Economy and Supply-Chain Stability:** Through its subsidiary Brunp, CATL has pioneered a closed-loop "production–use–recycling" model. Recovery rates for nickel, cobalt, and manganese have exceeded 99%, while lithium recovery surpasses 90% [29]. This integration mitigates reliance on virgin minerals and hedges against raw material price volatility.

4.2 Weaknesses (Internal Negative Factors)

- **High Revenue Concentration:** Power batteries account for over 70% of total revenue [26]. Despite rapid growth in Energy Storage Systems (ESS), performance remains highly sensitive to the global electric vehicle market cycle.
- **Domestic Market Dilution:** As competitors like BYD and CALB rise, automakers are adopting "multi-sourcing" strategies to ensure supply-chain security. Consequently, CATL's domestic market share has faced pressure, sliding from approximately 50% to 41.7% [30].
- **Global Compliance and Operational Costs:** The transition to a global enterprise has squeezed overseas margins. In Hungary and Germany, CATL faces stringent environmental regulations, complex labor laws, and cross-cultural management hurdles. Environmental disputes over water consumption in Hungary have incurred legal challenges and rising labor costs, weighing on short-term profitability [31].

4.3 Opportunities (External Positive Factors)

- **Sodium-ion Batteries as a "Second Growth Curve":** 2026 marks the first year of large-scale sodium-ion commercialization. By leveraging abundant sodium resources to hedge against lithium price swings, CATL aims to capture the energy storage and entry-level EV markets. A dual-track strategy (sodium-ion for cost-efficiency and solid-state for peak performance) secures long-term pricing power [32].
- **Strategic Boom in Energy Storage Systems (ESS):** ESS has emerged as a "Golden Sector" outperforming power batteries. Global shipments are expected to surpass 650 GWh by 2025, with a projected 40% growth in 2026 [33]. CATL's EnerX systems solve key customer pain points, establishing ESS as the company's second major growth engine.
- **Skateboard Chassis and Vertical Integration:** The CATL Integrated Intelligent Chassis (CIIC) integrates batteries, motors, and the chassis into a single unit. This business-model innovation redefines the relationship between suppliers and OEMs, deepening client lock-in [20].

4.4 Threats (External Negative Factors)

- **Geopolitics and Policy Barriers:** The U.S. FEOC (Foreign Entity of Concern) rules and the EU Battery Regulation impose significant hurdles for Chinese batteries entering North American and European markets [34].
- **Overcapacity and Price Wars:** Global planned capacity far exceeds actual demand, leading to declining utilization rates. With the 2024 industry average utilization at only 50%, aggressive price wars are severely compressing margins for both top-tier and second-tier players [19].

- **Disruptive Technological Risks:** Solid-state batteries represent the “ultimate” next-generation solution. While currently in pilot phases, competitors in Japan, South Korea (e.g., Toyota, Samsung SDI), and the West aim for low-cost mass production between 2027 and 2030 [35]. Any breakthrough by rivals could threaten CATL’s liquid-electrolyte dominance and render existing production lines obsolete.

5 Valuation: Fundamentals versus Market Pricing

CATL’s valuation metrics reveal a clear alignment between its intrinsic value and market positioning.

Capital Efficiency and Premium. A 24.36% ROE underscores exceptional capital utilization, fueling the company’s internal growth trajectory. Complementing this, a 7.06× PB ratio reflects the market’s willingness to pay a premium for CATL’s dominant technological moat and global scale to secure “alpha” returns.

Earnings Digestion and Growth Catalyst. The contraction from a 42.64 static PE to a 33.93 forward PE indicates that rapid EPS expansion is successfully “digesting” high valuation levels. This sets the stage for a “Davis Double Play”—where a recovery in market sentiment could trigger a powerful, simultaneous surge in both earnings and valuation multiples.

Cash Quality and Valuation Floor. A PCF ratio of approximately 14.99 confirms a high-quality conversion of profits into cash flow, providing a robust baseline for discounted-cash-flow (DCF) modeling. This fundamental strength ensures that CATL’s RMB 1.87 trillion market capitalization is anchored in financial reality rather than speculative sentiment.

Part II Investor Sentiment and Stock Pricing: An Empirical Study

6 Introduction

The new energy vehicle (NEV) industry chain has experienced a superimposed process of policy drive, technological iteration, and capital-market valuation reshaping over the past decade. As a global leader in the power battery industry, CATL’s (300750.SZ) stock price not only reflects the company’s own capacity expansion, customer structure, and earnings expectations, but also largely acts as a “weathervane” for investors observing the entire lithium-battery track. Against this background, public information surrounding CATL presents an obvious “dual-track” characteristic: on the one hand, research-report texts formed by sell-side brokers based on financial models and industrial research, whose emotional expression is relatively restrained and whose time distribution is highly coupled with earnings reports and policy nodes; on the other hand, massive posts in retail-investor forums such as East Money Guba, where sentiment fluctuates violently and reacts rapidly to hot events, but with greater noise and mixed calibers. Traditional asset-pricing research mostly focuses on financial statements, macroeconomic liquidity, or industry-prosperity indices. Although some recent literature has incorporated news sentiment and social-media texts into the stock-price explanation framework, empirical research directly comparing institutional channels and retail channels for the same target, and incorporating the price linkage of supply-chain competitors into the mechanism test, remains relatively insufficient—especially in the context of China’s A-share NEV leaders, a complete, reproducible chain of evidence bridging monthly and daily frequencies is still lacking.

Theoretically, there are at least three identifiable pathways through which investor sentiment affects stock prices. The first is the direct pricing pathway: optimistic or pessimistic sentiment alters investors’ risk preferences and trading willingness, thereby affecting the demand curve and short-term prices [36]. The second is the information-revelation pathway: sentiment is not purely noise, but rather a “collective guess”

regarding undisclosed fundamentals or policy expectations; its relationship with future returns may serve as a leading indicator, or it may be rapidly absorbed under efficient-market conditions. The third is the supply-chain spillover pathway: the stock prices of lithium-battery leaders are highly synchronized with those of downstream auto manufacturers and upstream material leaders. Sector-wide fluctuations alter market attention to the entire track, which is then transmitted to individual stocks through sentiment or capital allocation. In this case, sentiment may act as a mediating variable or merely as an accompanying phenomenon of price linkage. The particularity of the CATL case lies in the fact that it is both an important carrier of sector beta and one of the most actively discussed individual stocks by retail investors. If one relies on a single sentiment indicator or examines only the individual stock's own returns, it is easy to misjudge "sector resonance" as "sentiment causality," or mistake "retail noise" for "institutional expectation."

The research questions (RQs) of this paper are set as follows. RQ1 (dual-channel comparison): at the monthly frequency, between institutional research-report sentiment and retail Guba sentiment, which exhibits stronger concurrent and leading correlations with CATL's monthly returns? Do retail investors exhibit the "overreaction" style leading effect commonly seen in literature? RQ2 (supply-chain mediation): does the monthly return of a competitor portfolio have a significant total effect on CATL's returns? If so, is this effect significantly mediated by industry sentiment, retail sentiment, or institutional sentiment? RQ3 (daily-frequency prediction): in the sample of trading days where sentiment and price-volume data are aligned, can single-day sentiment or composite sentiment indicators (including persistence, momentum, relative competitor sentiment difference, etc.) provide distinguishable predictive information for the next day's up/down direction? RQ4 (feature ablation): under the Leave-One-Out (LOO) cross-validation framework, between sentiment features, price-volume technical features, and their combinations, which contributes more to classification accuracy? These four questions progress from shallow to deep, forming a progressive structure of "monthly mechanism → daily structure → machine-learning robustness."

The retail sentiment in this paper is mapped to the [0, 1] interval via SnowNLP using Guba titles and main texts, and then aggregated by day and month. The institutional sentiment is derived from research-report texts through a FinBERT/rating-fallback process to obtain monthly averages. The monthly-frequency section primarily employs Pearson correlation and Baron–Kenny stepwise-regression mediation analysis, estimating the indirect effects with the Sobel test and 1,500 bootstrap confidence intervals; the competitor-portfolio returns are constructed using equal-weighted monthly returns. In the daily-frequency section, over 122 valid trading days, 16-dimensional raw features and 18-dimensional enhanced features are constructed. Using the next day's positive or negative return as the label, six types of models—Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and Naive Bayes—are evaluated using LOO. An ablation experiment ("sentiment only / price-volume only / all features / composite sentiment") is conducted to compare information-set contributions. It should be noted that industry- and competitor-related sentiment in some months contains imputations based on a mix of regression from observed months and time interpolation. The main text reports both the observation-only sample and the imputed sample in parallel to avoid presenting inferred values as raw scraped conclusions. Retail sentiment in this sample has achieved full month-by-month post coverage from 2023 to 2026, meaning its monthly averages are mainly derived from real captures rather than imputation.

The marginal contributions of this paper are threefold. First, it proposes and empirically compares the explanatory power of institutional and retail dual-channel sentiment on the pricing of the same leading stock, finding that the linear correlation of both with returns at the monthly frequency is weak, thereby providing a cautious correction to the popular narrative that "social-media sentiment drives monthly

trends." Second, it incorporates supply-chain competitor returns, sentiment, and the leader's return into a unified mediation framework. Premised on significant total-effect covariance, it identifies whether sentiment acts as a statistically significant mediation channel, providing a testable articulation for the mechanism debate between "sector resonance" and "sentiment transmission." Third, it introduces a sentiment-persistence structure (such as consecutive high-sentiment days) at the daily frequency, indicating that the informational value of sentiment is more reflected in time-series patterns than in single-day cross-sectional levels. Through LOO machine learning, it confirms that sentiment features slightly outperform pure price-volume features out-of-sample, whereas a mix of all features paradoxically shows signs of overfitting, suggesting that information blocks should be separated rather than simply stacked when modeling.

The remainder of Part II is structured as follows: Section 7 reviews the literature and develops the research hypotheses; Section 8 introduces data sources, variable construction, and empirical strategies; Section 9 reports the monthly-frequency correlation and mediation-effect results; Section 10 analyzes daily-frequency composite sentiment and machine-learning predictions; Section 11 discusses robustness and limitations; and Section 12 concludes and proposes future research directions.

7 Literature Review and Research Hypotheses

7.1 Investor Sentiment and Asset Pricing: From Macro Indices to Textual Measures

Investor sentiment is a critical variable explaining asset-price deviations from fundamentals, cross-sectional return differences, and volatility clustering. Early research predominantly characterized overall market sentiment via indirect proxies such as survey indices, closed-end fund discounts, turnover rates, or IPO activity [37], [36], finding that high sentiment is often followed by lower subsequent market returns, especially for small-cap and highly speculative stocks. With the abundance of computable text resources, literature gradually shifted to direct textual sentiment: media news, corporate announcements, social-media posts, and sell-side research reports have all been proven to contain incremental information regarding future returns, volatility, or trading volume [38], [39]. Under the behavioral-finance framework, sentiment affects prices through two channels: first, by altering investors' risk tolerance and trading demand, causing prices to deviate from rational expectations in the short term; second, as a "collective interpretation" of incompletely priced information, presenting a testable lead-lag structure under weak efficient-market conditions.

However, whether sentiment is "useful" highly depends on measurement caliber, frequency, and sample period. For the same target, it may appear merely as noise at the monthly frequency but exhibit persistent structure at the daily frequency. A single aggregate sentiment index also struggles to distinguish information sources (professional institutional expression vs. emotional retail expression). Therefore, recent case studies tend to match sentiment variables with specific information channels, specific industry leaders, and specific frequencies, rather than solely reporting market-wide average effects. This paper follows this lineage, descending sentiment research from the "market level" to the individual-stock level while embedding the sector-linkage context of the NEV industry chain.

7.2 Sentiment Measurement Based on Chinese Texts: Guba, Research Reports, and NLP Methods

China's capital market features high retail participation, active online discussions, and concentrated sell-side coverage, providing rich materials for textual-sentiment research while also bringing measurement divergences. Investor-forum data from Guba, Xueqiu, etc., are widely used to characterize retail-investor

sentiment: early works mostly adopted sentiment dictionaries or lightweight tools like SnowNLP, finding that post sentiment correlates with trading volume, volatility, and short-term returns. However, the results are sensitive to extreme-value processing, duplicate-post filtering, and sentiment thresholds (relevant research mainly focuses on main-board consumption, finance, and individual hot-theme stocks). Sell-side research reports represent institutional information channels: the text is highly professional, and ratings and earnings forecasts possess structured fields. Sentiment can be extracted via pre-trained financial-domain models like FinBERT, or "buy/overweight" ratings can be mapped into ordinal sentiment scores. The two channels often present a sharp contrast in distribution—"narrow institutional range vs. wide retail fluctuation." If compared directly without alignment, it is easy to draw one-sided conclusions such as "retail is more predictive" or "institutions are more authoritative."

Methodologically, Chinese financial-text sentiment analysis faces challenges such as word-segmentation ambiguity, ironic expressions, clickbait titles, and announcement noise. Academia typically adopts three types of responses: (1) rule-based cleaning and deduplication to exclude low-information posts like reprinted announcements and risk warnings; (2) domain dictionaries or deep-learning models to improve financial-semantic resolution; (3) aggregating from daily to monthly or weekly frequencies to reduce micro-level noise. This paper maps retail data to the [0, 1] interval using SnowNLP and aggregates it daily and monthly; for the institutional side, it employs a FinBERT/rating-fallback process, testing separately on monthly and daily frequencies to address the "caliber-divergence" issue. It is worth pointing out that textual sentiment is not an exogenous variable; it has endogenous interactions with contemporaneous returns, trading volume, and media-coverage intensity. Therefore, this paper prioritizes reporting concurrent and lagged correlations at the monthly frequency, and utilizes next-day return labels and leave-one-out cross-validation at the daily frequency to mitigate sample-in overfitting critiques.

7.3 Institutional Sentiment versus Retail Sentiment: Theoretical Expectations of Dual-Channel Comparison

Institutions and retail investors exhibit systematic differences in information processing, trading constraints, and sentiment expression. Institutional investors typically have research teams, compliance constraints, and portfolio-management goals. Their public research-report sentiment is closer to a smooth assessment of fundamentals, and its impact on short-term prices may be weak, potentially even exhibiting lag due to "rating stickiness" [40]. Retail investors are more easily driven by narratives, policy rumors, and sector hot spots; their sentiment changes rapidly and carries high tail risk, theoretically more likely to reflect overreaction or attention-driven trading [41], [42].

In the case of China's NEV leading stocks, the dual-channel comparison carries special implications. As a target heavily held by institutions and deeply covered by the sell-side, CATL's research-report sentiment may already be partially reflected in industry comparisons and competitor pricing. Meanwhile, Guba discussions often surge during policy inflection points, capacity news, and periods of massive stock-price volatility; retail sentiment may be closer to the short-term trading level rather than monthly fundamentals. Consequently, while the literature generally acknowledges the differences between retail and institutional sentiment, there is no consensus on "which better explains the leader's returns at the monthly frequency": some studies find social-media sentiment has weak predictive power for short-term returns, while others indicate sentiment coefficients decay rapidly after controlling for price-volume and industry factors. This paper empirically establishes a comparable benchmark by testing concurrent and lagged correlations under overlapping-month samples (where both institutional and retail observations exist) and the full retail sample respectively.

7.4 Supply-Chain Linkage, Spillover Effects, and Sentiment Mediation

The new-energy battery industry exhibits significant supply-chain-resonance characteristics: vehicle-electrification expectations, upstream lithium-salt prices, competitors' capacity rhythms, and policy subsidies collectively shape the sector's beta. Research on spillover effects in financial economics shows that price changes in affiliated companies can be transmitted to the target company via information transfer, capital reallocation, or industry-index arbitrage [43], [44]. For CATL, price changes in targets like BYD (vehicles + self-supplied batteries), Ganfeng Lithium (upstream resources), and EVE Energy (peer battery manufacturer) may reflect common-factor shocks while also altering investors' beliefs about the leader's competitive advantages and market share.

Against this backdrop, whether sentiment constitutes a mediating variable becomes key to mechanism identification. According to Baron and Kenny [45] and subsequent mediation-analysis literature, if competitor returns affect the leader's returns entirely through sentiment transmission, the indirect effect should be significant, and the direct effect c' should be noticeably weakened relative to the total effect c. If sentiment is merely an "accompanying phenomenon" of price linkage, the total effect will be significant while the indirect effect will not. Existing industrial-finance research mostly remains at the "linkage description" or "Granger causality" level, rarely distinguishing among the three mediating pathways of industry sentiment, individual retail sentiment, and institutional sentiment within the same framework, nor do they often report Sobel statistics and bootstrap confidence intervals. This paper constructs equal-weighted monthly competitor returns as the independent variable, using CATL retail sentiment, institutional sentiment, and industry/competitor sentiment as mediators to test the "sector shock → sentiment → leader pricing" chain, while also appending a lagged specification to observe short-term transmission delays.

7.5 Daily-Frequency Sentiment Structure, Persistence Characteristics, and ML Prediction

Monthly-frequency analysis fails to capture intraday fermentation and multi-day accumulation of Guba sentiment. Behavioral experiments and high-frequency market-microstructure studies suggest that investor attention and sentiment shocks often exhibit persistence: consecutive optimistic posts may reinforce herd behavior, while consecutive pessimistic posts may amplify selling pressure. Therefore, expanding sentiment from "single-day cross-sectional levels" to "persistence structures" (e.g., consecutive days above a threshold, 5-day persistence ratio, momentum, and z-score) holds theoretical validity. Recent applications of machine learning in financial forecasting show that nonlinear models can mine weak signals amidst feature interactions, but they also face small-sample overfitting and data-snooping risks [46]. Leave-One-Out (LOO) cross-validation, as an extreme out-of-sample scheme, is well suited for scenarios with limited daily-frequency sample sizes, enabling relatively fair comparisons of the marginal contributions of sentiment features, price-volume technical features, and their combinations.

In this paper, a composite sentiment indicator sent_bull_index is constructed at the daily frequency, integrating dimensions such as consecutive high-sentiment days, 5-day persistence ratio, sentiment momentum, standardized scores, and relative competitor sentiment differences. An ablation experiment ("sentiment only / price-volume only / all features / composite sentiment") is established. In the literature, this design corresponds to two testable propositions: first, whether the composite structure outperforms the single-day SnowNLP score; second, whether sentiment features outperform traditional price-volume technical indicators out-of-sample. If sentiment only shows significant correlation in the full sample but LOO classification fails to beat the baseline, it supports the "weak information, prone to overfitting"

judgment. If persistence variables outperform single-day sentiment in both correlation and LOO, it supports the "structure over level" judgment.

7.6 Research Hypotheses

Based on the aforementioned literature and the research questions (RQ1–RQ4), the following hypotheses are proposed. Note: the hypotheses adopt the "directional statement + testable statistic" format; the empirical section uses Pearson correlation coefficients, OLS path coefficients, Sobel tests, and LOO classification accuracy as test statistics, with a significance level of $\alpha = 0.05$. Robustness and limitations are reported in Section 11. Table 1 maps each hypothesis group to its empirical section and main test statistic.

Table 1. Mapping of hypothesis groups to empirical sections and test statistics

Hypothesis Group	Code	Empirical Section	Main Statistic
Dual-channel monthly	H1a–H1d	Section 9	Pearson (r, p)
Supply-chain mediation	H2, H3a–H3d	Section 9	OLS (a, b, c, c′), Sobel, Bootstrap CI
Daily structure	H4a–H4c	Section 10	Pearson, grouped means
Machine learning	H5a–H5d	Section 10	LOO accuracy, ablation comparison

7.6.1 Monthly-Frequency Dual-Channel Sentiment Hypotheses (RQ1)

H1a (Institutional Concurrent Effect): CATL's monthly return rate is positively correlated with the institutional research-report sentiment of the current month, i.e., the more optimistic the sell-side sentiment, the higher the monthly return.

H1b (Retail Concurrent Effect): CATL's monthly return rate is positively correlated with the retail Guba sentiment of the current month, i.e., the more optimistic the forum sentiment, the higher the monthly return.

H1c (Retail Leading Effect): The previous month's retail Guba sentiment is positively correlated with the current month's return rate, and the correlation strength is higher than that of the previous month's institutional sentiment on the current month's return, reflecting the short-term leading nature of retail sentiment.

H1d (Dual-Channel Divergence): Institutional sentiment and retail sentiment are significantly positively correlated in overlapping months; if significance is insufficient, it indicates that the information content of the two channels has diverged rather than simply being redundant.

7.6.2 Supply-Chain Spillover and Mediation Hypotheses (RQ2)

H2 (Sector Resonance — Total Effect): The total effect c of the competitor portfolio's monthly return on CATL's monthly return is significantly positive, i.e., there is significant directional linkage between competitor and leader returns.

H3a (Industry Sentiment Mediation): The path coefficient a of competitor returns on industry/competitor sentiment is significant, and the path coefficient b of sentiment on CATL's return is significant after controlling for competitor returns; the indirect effect $a \cdot b$ is significantly different from zero according to the Sobel test.

H3b (Retail Sentiment Mediation): The indirect effect of Competitor Return → CATL Retail Sentiment → CATL Return is significant, reflecting that sector shocks transmit to leader pricing through retail discussions.

H3c (Institutional Sentiment Mediation): The indirect effect of Competitor Return → CATL Institutional Sentiment → CATL Return is significant, reflecting that sector shocks transmit to leader pricing through sell-side expectation adjustments.

H3d (Delayed Mediation): The indirect effect of the lagged one-period competitor return on the current-period CATL return, mediated through industry sentiment, is significant, reflecting the short-term delay of sentiment transmission.

7.6.3 Daily-Frequency Sentiment Structure and Prediction Hypotheses (RQ3)

H4a (Single-Day Sentiment Weak Prediction): The daily SnowNLP sentiment (nd_sentiment) is significantly positively correlated with the next day's return rate.

H4b (Structure over Level): The correlation strength of the composite sentiment indicator (sent_bull_index) and consecutive high-sentiment days with the next day's return is higher than that of single-day sentiment; grouped by consecutive high days, the high-persistence group's average next-day return is significantly higher than the low-persistence group's.

H4c (Non-linear Grouping): Under the tertile grouping of sent_bull_index, there is a significant difference in the next day's average return between the high-sentiment group and the low-sentiment group, supporting a monotonic or non-monotonic predictable structure of sentiment on returns.

7.6.4 Machine-Learning Ablation Hypotheses (RQ4)

H5a (Sentiment Out-of-Sample Information): Under the LOO framework, the best classification accuracy of the sentiment-only feature set is significantly higher than the random baseline (majority-class proportion of next day's up/down).

H5b (Price-Volume Out-of-Sample Information): The best LOO accuracy of the price-volume technical feature set alone is significantly higher than the random baseline.

H5c (Sentiment vs. Price-Volume): The best LOO accuracy of the sentiment-only feature set is higher than the price-volume-only feature set, indicating that sentiment features provide marginal contributions to short-term direction prediction.

H5d (Composite Enhancement): The best LOO accuracy of the enhanced feature set (composite sentiment + technical features) is higher than that of the original 16-dimensional full feature set, indicating that persistent-sentiment construction mitigates noise and improves out-of-sample performance.

8 Data and Research Design

8.1 Sample Period and Data Sources

This paper takes CATL (stock code 300750.SZ) as the research subject, with the sample period covering January 2023 to June 2026. This period encompasses multiple phases such as NEV-sector policy adjustments, lithium-price fluctuations, capacity-release expectations, and massive stock-price volatility, which is conducive to testing the sentiment–return relationship against a relatively unified industry backdrop. The data consists of four types of sources: (1) retail-sentiment raw texts, derived from titles and bodies of East Money Guba posts; (2) institutional-sentiment raw texts, derived from sell-side research-report abstracts or full-text snippets; (3) individual-stock and competitor market data, derived from East Money daily opening price, closing price, trading volume, etc.; (4) daily-frequency machine-learning panel, generated by aligning the aforementioned sentiment and price-volume features by trading day. The stock-price interval for the main analysis ends in April 2026 (40 trading months in the monthly panel). The daily-

frequency prediction sample consists of 122 trading days with non-missing sentiment and price-volume data, and the enhanced composite-feature subsample consists of 98 trading days. Table 2 summarizes the data sources and sample sizes.

Table 2. Overview of data sources and sample sizes

Data Type	Source	Raw Sample	Aggregation Level
Retail Guba text	East Money Guba	28,929 posts	Daily 214 days / Monthly 42 months
Institutional report text	Brokerage DB	462 reports	Monthly 27 months (with stock-price overlap)
CATL stock data	East Money	Daily data	Monthly close, 40 months
Competitor market data	East Money	Daily data	Equal-weight monthly return
Daily ML panel	Program generated	122 days	Next-day up/down label

The collection of Guba data employed time-sorted pagination scraping. Posts were deduplicated (using Post_Id as the primary key), empty titles and reprinted announcements (containing keywords like "announcement," "information disclosure") were excluded, retaining valid discussion posts published within the research period. Verification showed that retail sentiment at the monthly frequency has achieved full coverage for 42 calendar months; thus, monthly retail-sentiment averages are primarily based on real observations rather than imputation. Institutional research-report sentiment is restricted by report-publishing rhythms, yielding an effective overlap of 28 months with stock-price monthly returns.

8.2 Variable Definition and Construction

8.2.1 Dependent Variables

(1) Monthly return rate (monthly_return_pct). Aggregated from CATL's daily closing prices by calendar month: taking the opening price of the first trading day and the closing price of the last trading day of each month. Both monthly correlation analysis and mediation analysis use this as the primary dependent variable.

(2) Next-day up/down direction (target). Defining R(t+1) as the return rate of the t+1 trading day relative to the t-th day. The continuous return indicator next_day_return is used for Pearson correlations and grouped-mean comparisons.

8.2.2 Core Independent Variables: Dual-Channel Sentiment

Retail sentiment (retail_sentiment_mean). For each Guba text, SnowNLP is used to calculate the sentiment score and linearly mapped to the [0, 1] interval (0 tending to negative, 1 tending to positive). At the daily-frequency level, the arithmetic average of all post scores on the same trading day is taken, denoted as nd_sentiment; at the monthly-frequency level, the average of all post scores for that month is taken, resulting in CATL's monthly retail average. To test the leading effect, a one-period lagged variable retail_sentiment_lag1 is constructed.

Institutional sentiment (inst_sentiment_mean). For each research-report text, the Chinese financial-sentiment model IDEA-CCNL/Erlangshen-Roberta-110M-Sentiment (FinBERT-like) is prioritized to output sentiment scores. When the model is unavailable or the text is too short, the sell-side rating (buy/overweight/neutral/underweight/sell) is mapped to [−1, 1] and then shifted to [0, 1] as a fallback (--use-rating-fallback). The arithmetic average of research-report sentiment scores is taken monthly.

8.2.3 Supply-Chain and Competitor Variables

The competitor portfolio includes BYD (002594), Ganfeng Lithium (002460), and EVE Energy (300014), which together with CATL constitute a representative sample of the lithium-battery industry chain. The following variables are constructed:

- Competitor equal-weighted monthly return (peer_return_mean): the cross-sectional arithmetic mean of the monthly return rates of the three competitor stocks, serving as the core independent variable (X) in the mediation model.
- Competitor return volatility (peer_return_std): the standard deviation of the three competitors' returns in the same month, used to characterize intra-sector divergence (auxiliary feature for imputation models).
- Industry/competitor sentiment (peer_sentiment_mean): for months where Guba sentiment observations exist for the three competitors, the SnowNLP monthly average is taken; missing months are imputed according to the rules in Section 8.3. In the daily-frequency analysis, the CATL–BYD sentiment difference (diff_byd) and its lagged terms, 5-day averages, and duration days are constructed to depict the relative narrative strength between the battery leader and the vehicle-manufacturer leader.

8.2.4 Daily-Frequency Price-Volume and Technical Features

Based on data obtained from East Money, the following price-volume variables are included in the daily-frequency panel: return (daily return), return_1/3/5 (1/3/5-day cumulative returns), return_lag1/2 (lagged returns), MA_5_ratio and MA_10_ratio (ratio of closing price to 5/10-day moving averages), volume_ratio (ratio of trading volume to 5-day moving-average volume), volatility_5 (5-day return volatility), and relative_strength (relative-strength index). The above variables and sentiment variables are aligned on the same trading day; rows with missing values are removed prior to ML estimation.

8.2.5 Composite Sentiment Indicator (Daily-Frequency Enhancement)

Given the large noise in single-day SnowNLP data, this paper constructs 7-dimensional composite sentiment features and a comprehensive index sent_bull_index based on nd_sentiment and diff_byd, characterizing the persistence, momentum, and extreme states of sentiment. The enhanced feature set (18 dimensions) = 7-dimensional composite sentiment + 11-dimensional price-volume technical features.

8.3 Missing-Value Handling and Imputation Strategy

Retail sentiment. Through targeted supplementary scraping from 2023 to 2026, the CATL Guba has post records for all 42 calendar months. The monthly-frequency retail sentiment can be directly aggregated from

observed posts without statistical imputation for retail monthly averages. It should be noted that the number of posts is sparse in some months (e.g., only 11 posts in May 2025), and the estimation precision of the monthly average remains lower than in post-dense months (e.g., 7,916 posts in March 2023). Section 9 balances coverage completeness and post-density heterogeneity when interpreting the results.

Competitor-industry sentiment and institutional sentiment. Competitor Guba sentiment is missing in some months. The following methods are used for supplementation: (1) time linear interpolation—linear interpolation is performed on the sentiment series across observed months to obtain the interp component; (2) multiple-regression prediction—estimated on observed months, generating regression predictions reg_pred for the full-sample months; (3) mixture—for missing months, 45% interpolation + 55% regression prediction is applied; for observed months, the original values are retained.

8.4 Machine Learning and Ablation Experiments (Testing H5)

Table 3 defines the four feature sets used in the LOO ablation experiments.

Table 3. Feature sets for the machine-learning ablation experiment

Code	Feature Set	Dimensions	Purpose
Full	Raw 5-dim sentiment + 11-dim price-volume	16	Full-feature baseline
Sent	Only raw sentiment	5	Test H5a, H5c
Tech	Only price-volume technical	11	Test H5b, H5c
Enhanced	Composite 7-dim sentiment + 11-dim price-volume	18	Test H5d

9 Empirical Results: Monthly Frequency

This section reports the hypothesis-testing results at the monthly frequency, corresponding to H1 (dual-channel sentiment correlation) and H2–H3 (supply-chain spillover and mediation effects) proposed in Section 7. The estimation sample is a monthly panel from January 2023 to April 2026, totaling 40 trading months. Among them, the overlapping months where both institutional research-report sentiment and retail Guba sentiment are observed amount to 28 months (the effective correlation-regression sample slightly reduces to 27 months due to lags and missing data). Retail sentiment has achieved full coverage of 42 calendar months from January 2023 to June 2026, and the monthly-frequency retail series in the correlation analysis is extended to 39 months (the subsample aligned with monthly returns).

9.1 Descriptive Statistics and Sample Characteristics

There are 28,929 original Guba texts, which form a retail-sentiment series of 214 daily frequencies and 42 monthly frequencies after mapping by SnowNLP. Post-volume distribution is highly uneven across months: reaching 7,916 posts in March 2023, while only 11 posts in May 2025, suggesting heterogeneity in the estimation precision of monthly averages. On the institutional side, a total of 462 research reports entered the sentiment scoring, resulting in 27–28 valid overlapping months with stock-price monthly returns. Figure 1 compares CATL's normalized monthly stock price with institutional and retail sentiment, while Figure 2 contrasts the two sentiment channels directly.

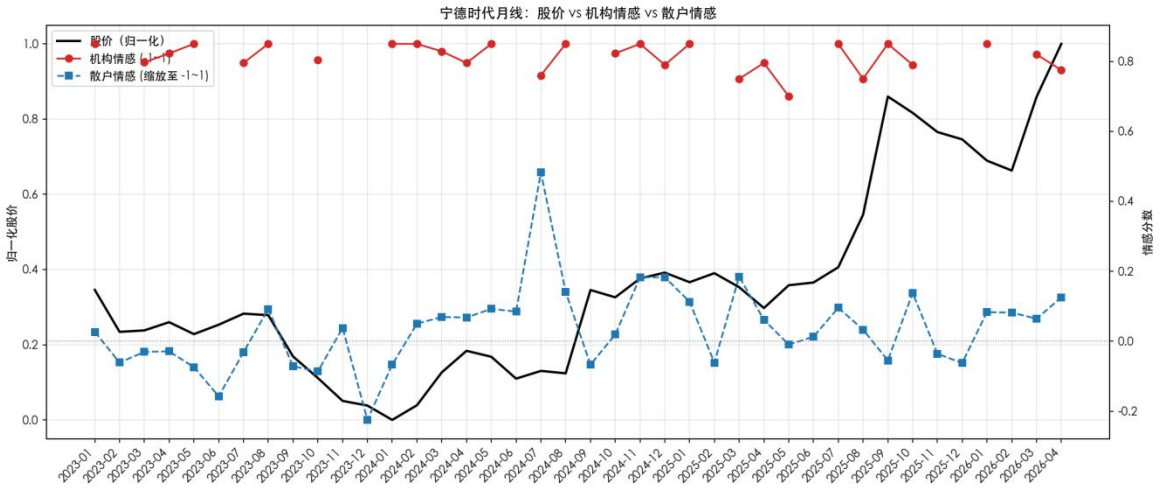


Figure 1. Three-line comparison of CATL monthly return, institutional sentiment, and retail sentiment

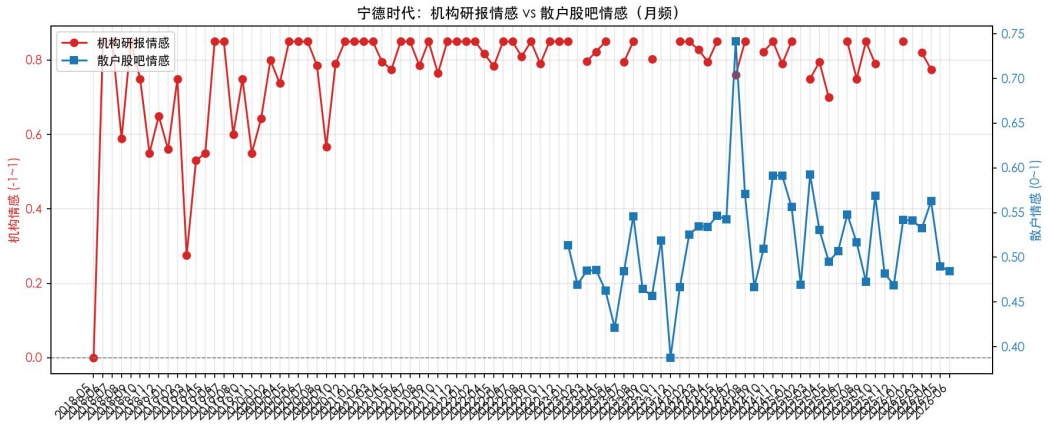


Figure 2. Scatter-plot comparison of dual-channel sentiment

It can be intuitively seen that institutional sentiment fluctuates within a narrow range of [0.4, 0.7], whereas retail sentiment has a larger amplitude, and neither exhibits a clear linear co-directional trend with monthly returns—this is consistent with the null results of the subsequent correlation tests.

9.2 Monthly-Frequency Correlation Analysis of Dual-Channel Sentiment (H1)

Table 4 summarizes the Pearson correlations of institutional and retail sentiment with CATL's current-month return rate, including both concurrent and one-period leading specifications.

Table 4. Monthly-frequency correlations of dual-channel sentiment with CATL returns

Variables	r	p value	n	Hypothesis	Conclusion
Institutional sentiment ~ current-month return (overlap)	−0.085	0.673	27	H1a	Not supported
Retail sentiment ~ current-month return (overlap)	−0.029	0.887	27	H1b	Not supported

Variables	r	p value	n	Hypothesis	Conclusion
Retail sentiment ~ current-month return (retail months)	0.019	0.907	39	H1b	Not supported
Last-month inst. sentiment ~ current-month return (lead)	0.086	0.744	17	H1c	Not supported
Last-month retail sentiment ~ current-month return (lead)	0.140	0.486	27	H1c	Not supported
Institutional sentiment ~ retail sentiment (overlap)	−0.213	0.277	28	H1d	Not sig., weak neg.

In the institutional–retail overlap sample (n = 27), the correlation coefficient between institutional sentiment and current-month return is (r = −0.085, p = 0.673), and between retail sentiment and current-month return is (r = −0.029, p = 0.887). Both are insignificant at the 5% level, and the direction is inconsistent with the positive correlation expected by H1a and H1b. After expanding the retail sample to all 39 months with Guba data, the correlation coefficient between retail sentiment and current-month return is only (r = 0.019, p = 0.907), leaving the conclusion unchanged. The above results indicate a rejection of the hypothesis that "institutional or retail sentiment is significantly positively correlated with the current month's return" at the monthly frequency (H1a, H1b). Institutional research-report sentiment possesses almost zero linear explanatory power for monthly ups and downs, which is consistent with characteristics like sell-side rating stickiness and excessively small sentiment-score variance; although retail sentiment fluctuates significantly at the daily frequency, its correlation with monthly returns is diluted by noise after aggregation.

In the leading one-period test, the correlation coefficient between last month's retail sentiment and current month's return is (r = 0.153, n = 39, p = 0.354), slightly higher than that of last month's institutional sentiment (r = 0.086, n = 17, p = 0.744). The direction partially aligns with the "retail leading" articulation of H1c, but neither reaches conventional significance levels. Thus, H1c cannot be statistically supported and can only be noted as an exploratory finding in discussion. The concurrent correlation between institutional and retail sentiment is (r = −0.213, p = 0.277, n = 28), showing a weak but insignificant negative correlation. This indicates that the dual channels are not merely redundant, nor do they form a significant consensus expectation (H1d does not support significant positive correlation, but supports the descriptive judgment of "divergence rather than overlap").

Summary (H1): in the monthly-frequency dimension, both the institutional and retail channels lack a significant concurrent or leading linear relationship with CATL's monthly returns. This suggests that neither Guba monthly averages nor research-report monthly averages should be solely used as core explanatory variables for monthly returns; instead, the identification strategy should shift towards supply-chain linkages and daily-frequency sentiment structures (see Section 10).

9.3 Supply-Chain Spillovers and Mediation Effects (H2–H3)

Before testing sentiment mediation, we first examine the total effect of the competitor portfolio's monthly return on CATL's monthly return. Table 5 shows that the total-effect coefficient c under the four path settings in the concurrent specification is uniformly 0.8419 (n = 39), and the Pearson correlation between competitor return and CATL return is as high as ($r = 0.808$, $p < 0.01$). In the lagged one-period specification, the total effect drops to ($c = 0.0869$, $n = 38$), and the contemporaneous correlation also weakens to ($r = 0.086$), which aligns with the empirical feature that "sector shocks are predominantly synchronous covariance, with weak lagged transmission." Consequently, H2 is supported: the equal-weighted portfolio returns of BYD, Ganfeng Lithium, and EVE Energy exert a strong synchronous positive linkage on CATL's monthly return. The lithium-battery leader and competitor stock prices share significant sector-beta components.

Given a significant total effect, we further test whether sentiment constitutes a statistically significant mediating channel. Table 5 reports the Baron–Kenny path coefficients, indirect effects (a·b), Sobel p values, and bootstrap 95% confidence intervals.

Table 5. Baron–Kenny mediation paths, indirect effects, and Sobel tests

Path	a (X→M)	b (M→Y|X)	c (total)	c′ (direct)	a·b (indirect)	Sobel p	Indirect 95% CI	Mediation ratio
A Peer → Industry sent. → CATL	0.0041	−11.0546	0.8419	0.8872	−0.0454	0.232	[−0.133, 0.010]	−5.4%
B Peer → Retail sent. → CATL	−0.0006	21.5069	0.8419	0.8556	−0.0137	0.543	[−0.055, 0.024]	−1.6%
C Peer → Inst. sent. → CATL	−0.0001	−2.3418	0.8419	0.8416	0.0003	0.943	[−0.025, 0.052]	~0%
D Lag 1 mo.: Peer → Industry sent. → CATL	0.0041	−0.3864	0.0869	0.0885	−0.0016	0.969	[−0.079, 0.105]	−1.8%

In Path A, the path coefficient of competitor returns on industry sentiment ($a = 0.0041$) is positive, but the conditional coefficient of sentiment on CATL return ($b = -11.05$) is negative. The indirect effect is ($a \cdot b = -0.0454$), Sobel ($p = 0.232$), and the bootstrap interval contains 0. H3a is not supported. For Path B (retail-sentiment mediation), the indirect effect is −0.0137, Sobel ($p = 0.543$), so H3b is not supported; sensitivity analysis using only observed retail months yields perfectly identical coefficients ($n = 39$) for Path B, leaving the conclusion unchanged. For Path C (institutional-sentiment mediation), the indirect effect is near 0 (0.0003), Sobel ($p = 0.943$), so H3c is not supported. For the lagged one-period Path D, Sobel ($p = 0.969$), and H3d is also not supported.

Auxiliary correlations further reveal the reasons for insignificant mediation: the correlation (r = 0.242) between competitor returns and industry sentiment is only moderate-to-weak, while the concurrent correlation between industry sentiment and CATL returns is merely (r = 0.027), nearly zero; the concurrent correlation between retail sentiment and CATL returns is also merely (r = 0.019). In other words, although sector price shocks are strong, they do not concurrently translate into an identifiable sentiment–return chain. The direct effect c′ (0.84–0.89) is almost equal to the total effect c, and the marginal contribution of sentiment variables is statistically negligible. These effects and the underlying sector linkage are visualized in Figures 3 and 4.

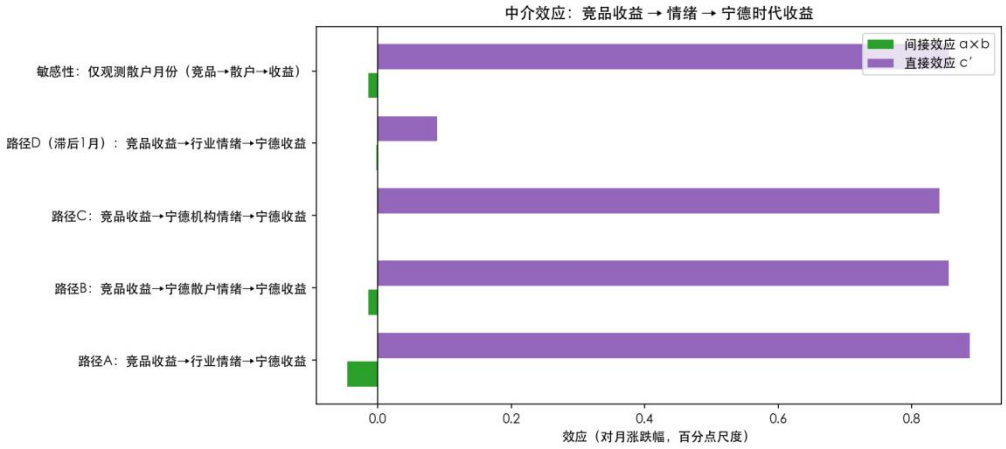


Figure 3. Total, direct, and indirect effects across the four mediation paths

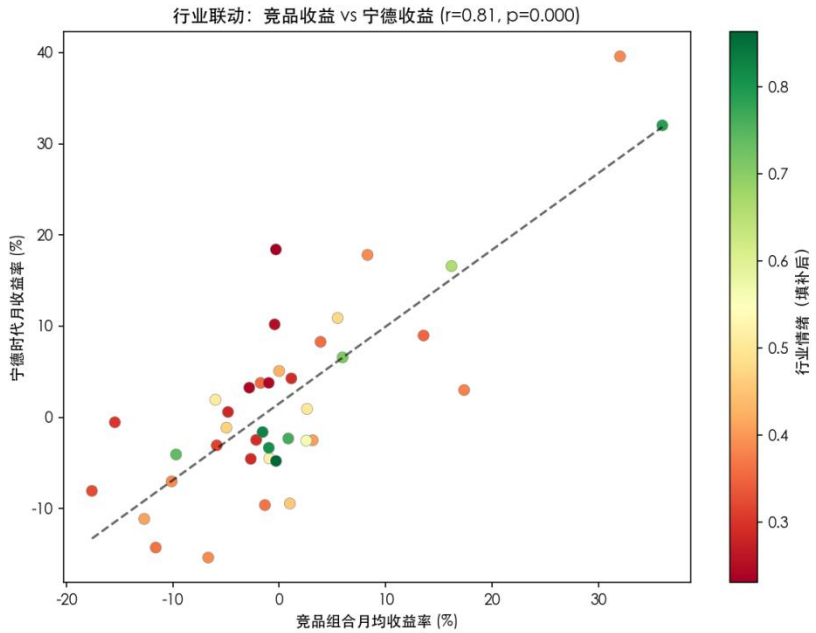


Figure 4. Sector linkage: competitor-portfolio return versus CATL return (colored by imputed industry sentiment)

Combined with Table 5, H2–H3 can be summarized as follows: (1) H2 holds—strong linkage between competitor and CATL monthly returns (c = 0.84, r = 0.81), supporting the "NEV lithium-battery sector-resonance" articulation. (2) H3a–H3d do not hold—whether through industry sentiment, retail sentiment, or institutional sentiment, indirect effects are uniformly insignificant, with Sobel p values ranging from a minimum of 0.232 to a maximum of 0.969. (3) Mechanism implication—the impact of competitor stock

prices on CATL stock prices is predominantly realized through direct price linkage or common-factor shocks, rather than mediated via observable textual-sentiment channels.

9.4 Summary of Monthly Empirical Evidence

Tables 4 and 5 jointly sketch a clear monthly-frequency picture: sector beta is significant; sentiment alpha is insignificant. Neither institutional nor retail dual channels can independently explain CATL's monthly fluctuations. Although competitor stock prices are highly co-directional with CATL, sentiment fails to constitute a reliable mediation mechanism. This indicates: (1) monthly aggregation obscures daily-frequency persistence structures; (2) linear correlation is insufficient to capture non-linear Guba outbursts; (3) the institutional rating channel itself may inherently not be a primary pricing factor for short-term monthly returns.

Therefore, Section 10 pivots to the daily-frequency sample (122 trading days) to test composite sentiment indicators and machine-learning out-of-sample predictions, responding to H4–H5 proposed in Section 7, and supplementing the mechanism queries left by the monthly-frequency null results from a higher-frequency perspective. The monthly empirics also highlight writing boundaries: even though retail Guba achieved full coverage over 42 months, monthly averages may still suffer compromised estimation quality due to uneven post densities (e.g., 11 posts in May 2025). Thus, daily and monthly conclusions should be reported in parallel and cross-verify one another, rather than acting as substitutes.

10 Daily-Frequency Sentiment Structure and Machine-Learning Prediction

Section 9 showed that neither institutional nor retail sentiment can significantly explain CATL's monthly returns at the monthly frequency, and supply-chain price linkages failed to form significant mediation via sentiment. This section pivots to the daily frequency, testing H4 (sentiment structure and next-day return) and H5 (ML out-of-sample prediction and feature ablation) on the trading-day sample where sentiment and price-volume are aligned. The valid sample consists of 122 trading days (2023-02-22 to 2026-04-22); upon introducing the composite-sentiment rolling features, the non-missing enhanced sample spans 98 days. The dependent variable is the next day's up/down direction (up = 1) or continuous return rate. Features are standardized prior to evaluating six classifiers via Leave-One-Out (LOO) cross-validation.

10.1 Motivation: From "Level" to "Structure"

Monthly-frequency null results do not imply that Guba texts contain no information. It more likely reflects: (1) single-day SnowNLP noise is averaged out within monthly means; (2) linear correlation cannot capture non-linear structures such as "multi-day consecutive bullishness" or "sentiment persistently strengthening relative to competitors." Therefore, this section first reports the Pearson correlations between single-day sentiment, composite indicators, persistence variables, and the next day's return (Table 6). Then it compares the next-day average returns grouped by consecutive high days (Table 7) and sent_bull_index tertiles (Table 8). Finally, it presents LOO classification and ablation results (Table 9).

10.2 Linear Correlation Tests (H4a–H4b)

Table 6. Correlations of daily sentiment metrics with next-day return (n = 97)

Indicator	r	p value	n	Hypothesis	Conclusion
Single day	0.148	0.147	97	H4a	10% level not significant

Indicator	r	p value	n	Hypothesis	Conclusion
Composite	0.159	0.119	97	H4b	Slightly better than single day, still insignificant
Sentiment momentum	0.040	0.694	97	—	Extremely weak
Consecutive high days	0.196	0.054	97	H4b	Marginally sig. at 10%

Table 6 summarizes the concurrent correlation between four sentiment metrics and next-day returns (n = 97). The correlation coefficient between single-day sentiment and next-day return is (r = 0.148, p = 0.147), which does not support H4a's significant positive correlation at the 5% level, showing only a weak positive trend at the 15% level. The composite index has (r = 0.159, p = 0.119), slightly higher than the single-day metric, aligning directionally with H4b's expectation of "structure over level," yet statistical significance remains insufficient. Notably, the correlation between consecutive high days and next-day returns reaches (r = 0.196, p = 0.054), marginally significant at the 10% level, providing preliminary evidence for the "persistence mechanism." Sentiment momentum shows virtually no linear association (r = 0.040, p = 0.694).

10.3 Non-linear Grouping: Persistence Rather than Monotonic Levels (H4b–H4c)

To alleviate the constraints of linear assumptions, the sample is divided into four groups based on consecutive high days to compare next-day average returns (Table 7), and into tertiles of sent_bull_index (Table 8); both groupings are visualized in Figure 5.

Table 7. Next-day average return grouped by consecutive high-sentiment days

Consecutive High Days	Next-Day Avg. Return (%)	n	Interpretation
0 days	−0.037	68	Baseline group
1–2 days	−0.610	17	Short-term high brings no positive return
3–5 days	+1.611	10	Persistent optimism accompanied by high return
6 days+	+1.400	2	Extremely small sample, reference only

Grouping results exhibit non-monotonic features: compared to the 0-day baseline group (−0.037%), the next-day average return of the 3–5 consecutive-high-days group surges to +1.611%, an economically meaningful magnitude. However, the 1–2-day group sees a drop to −0.610%, demonstrating that "brief highs" and "persistent highs" affect returns in opposite directions. This pattern is consistent with the behavioral-finance narrative of "attention gradually accumulating, consensus laggingly forming": single- or two-day sentiment spikes may accompany news realization or overreaction, while three to five days of persistent bullishness is more likely to correspond to stable fermentation of discussion heat. Due to only 2 observations in the 6+ days group, no inferences are drawn. Overall, H4b is partially supported—persistence structure is more informative than single-day levels, but the effect is threshold-based rather than simple monotonic.

Table 8. Next-day average return grouped by sent_bull_index tertile

Tertile	Next-Day Avg. Return (%)	n
Low	−0.300	32
Medium	+0.446	32
High	+0.042	33

The tertile grouping does not present a monotonic "High > Medium > Low" order: the medium-sentiment group sees the highest return (+0.446%), while the high-sentiment group is close to zero (+0.042%). The monotonic non-linearity expected by H4c is not supported. One explanation is that extremely high composite sentiment is already partially reflected in same-day prices, prompting next-day mean reversion. Another explanation is that limited sample sizes (approx. 32 days per group) lead to large volatility in group means.

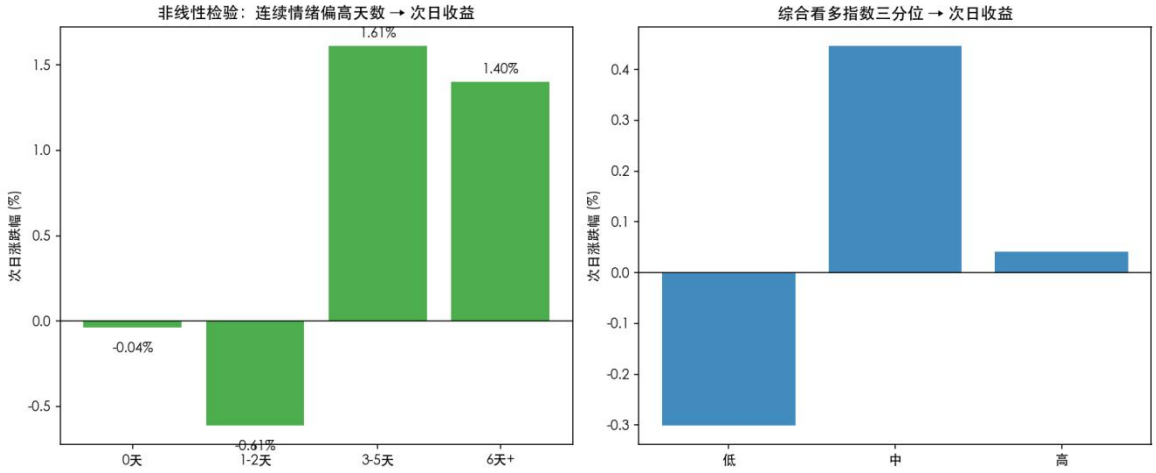


Figure 5. Next-day return grouped by consecutive high-sentiment days (left) and by composite-index tertile (right)

10.4 Machine-Learning Prediction and Feature Ablation (H5)

Setting the next day's up/down direction as the label, six types of models are executed via LOO. Each feature set takes the highest LOO accuracy among the six models as its representative, comparing it with the baseline accuracy of "always predicting the majority class" (Table 9).

Table 9. LOO classification accuracy across feature sets

Feature Set	Dims	Sample N	Best Model	LOO Accuracy	Baseline	Hypothesis	Conclusion
All features (raw)	16	122	Naive Bayes	0.549	0.557	—	Below baseline
Sentiment only (raw 5-dim)	5	122	Decision Tree	0.607	0.557	H5a, H5c	Above baseline
Price-volume tech only	11	122	Decision Tree	0.598	0.557	H5b	Above baseline

Feature Set	Dims	Sample N	Best Model	LOO Accuracy	Baseline	Hypothesis	Conclusion
Enhanced (comp. 7 + tech 11)	18	98	Logistic	0.582	0.571	H5d	Above baseline
Composite sentiment only	7	98	Naive Bayes	0.602	0.571	Suppl.	Above baseline

Main findings are as follows. First, "raw 5-dim sentiment + Decision Tree" achieves a 60.7% LOO accuracy, roughly 5 percentage points above the 55.7% baseline, with an F1 score of 0.455, supporting H5a (sentiment features contain out-of-sample information). Second, "11-dim technical features + Decision Tree" yields 59.8%, also above baseline, supporting H5b. Third, comparing the two, the sentiment set (60.7%) is slightly higher than the price-volume set (59.8%) by about 0.9 percentage points, granting weak support to H5c—sentiment offers marginal contributions comparable to or slightly better than price-volume in short-term directional prediction, though the advantage is limited. Fourth, the highest accuracy for the 16-dim all-feature mix is only 54.9%, underperforming the baseline. This hints that high-dimensional mixing easily overfits small samples, advising against simply stacking sentiment and price-volume features in one model. Fifth, the 18-dim enhanced set on the 98-day sample hits 58.2% (Logistic), above its 57.1% baseline, and visibly beats the 16-dim full-feature's 54.9%, supporting H5d—replacing raw single-day sentiment with persistent composite sentiment aids out-of-sample performance. The 7-dim composite sentiment + Naive Bayes reaches 60.2%, close to raw sentiment's 60.7% (Decision Tree), suggesting composite structures don't overturn conclusions but provide robust supplements across different model families. Figure 6 compares the LOO accuracy of the six models across feature sets, and Figure 7 reports the Random Forest feature importance on the enhanced set.

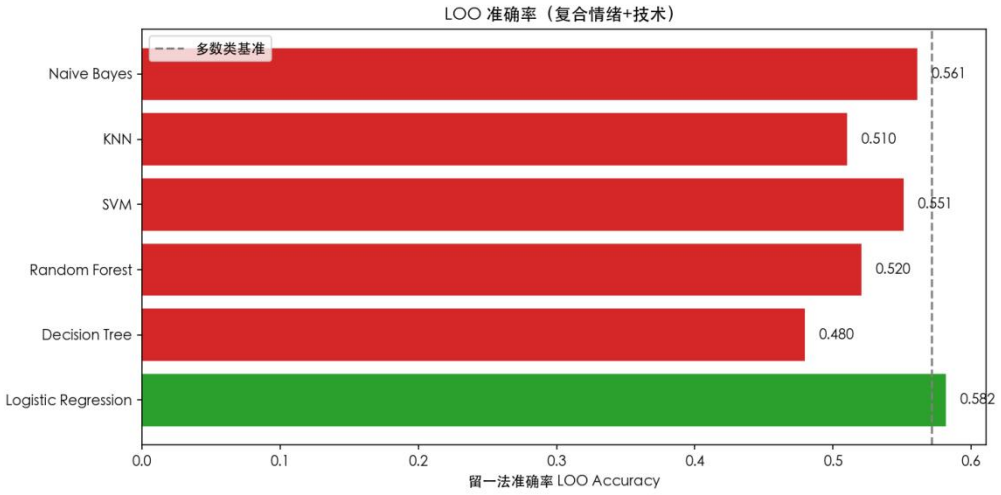


Figure 6. LOO accuracy of six models across feature sets

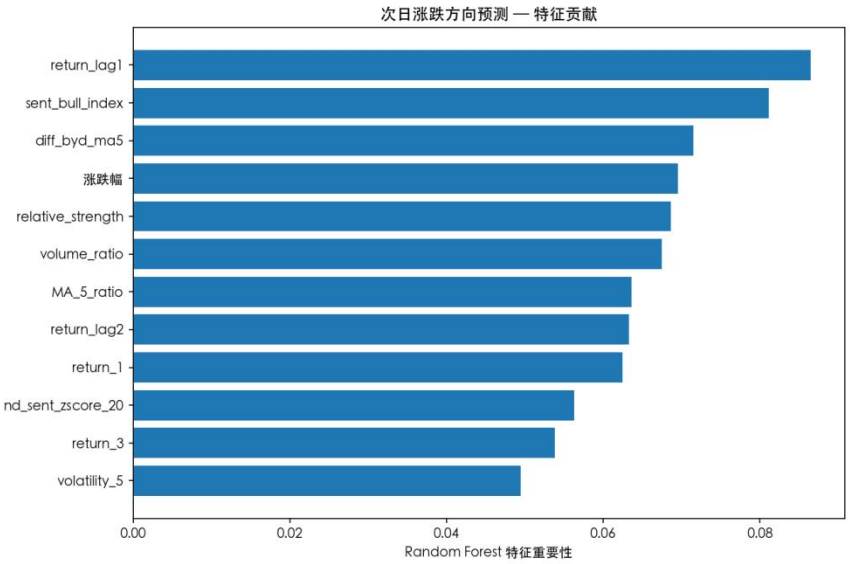


Figure 7. Random Forest feature importance on the enhanced feature set

On the 18-dim enhanced features, the top three Random Forest importances are: return_lag1 (0.086), sent_bull_index (0.081), and diff_byd_ma5 (0.071). Lagged returns reflect short-term price inertia, the composite bullish-sentiment index reflects persistent discussion structure, and the 5-day sentiment gap against BYD captures industry-chain comparative narratives. Together, these indicate that next-day directional prediction draws upon a mix of price-volume inertia, sentiment structure, and relative-track sentiment, rather than being dominated by any single variable.

10.5 Summary of Daily-Frequency Empirical Evidence

The daily-frequency results vividly contrast with the monthly-frequency results of Section 9 and can be summarized in three points.

First, frequency heterogeneity: while monthly correlations are insignificant, daily sentiment persistence (consecutive high days) reaches marginal 10% significance, and the 3–5-days group's next-day return outperforms the 0-day group by ~1.65 percentage points. This indicates sentiment information mostly manifests in daily dynamic structures rather than monthly averages.

Second, linear vs. non-linear: both single-day SnowNLP and the composite index show weak Pearson correlations ($r < 0.16$, $p > 0.10$), but grouping tests unveil threshold-based persistent effects; sent_bull_index tertiles exhibit no monotonic relationship, suggesting composite indicators are better suited for classification prediction than simple linear extrapolation.

Third, prediction and practical meaning: maximum LOO accuracy caps around 60.7% (barely 5% above baseline), lacking robust trading value but proving weak incremental out-of-sample information; the all-feature mix actually underperforms the baseline.

11 Robustness and Limitations

11.1 Robustness Discussion

This paper incorporates several endogeneity controls and sensitivity arrangements. While insufficient to perfectly identify causality, they mitigate obvious misjudgment risks.

(1) Dual-sample caliber comparison: monthly correlations simultaneously report the institutional–retail overlap period (n = 27–28) and the expanded retail sample (n = 39). Both yield consistent conclusions: neither concurrent nor leading correlations are significant. The institutional full sample (n = 27) matches the overlap results (r = −0.085), proving conclusions aren't driven by sample accidents.
(2) Mediation paths and lagged settings: beyond concurrent Baron–Kenny specifications, Path D lags competitor returns by one period. Total effect shrinks from 0.84 to 0.09, and indirect effects remain insignificant (Sobel p = 0.969), showing sentiment mediation lacks support across lagged dimensions as well.
(3) Imputation transparency: missing industry/competitor sentiments are filled via a "45% interpolation + 55% regression" mix; however, retail monthly frequencies originate from 42 months of true observations. Path B mediation rests strictly on observed retail series, untainted by imputation.
(4) Out-of-sample prediction: daily ML uses LOO instead of random splits to bypass optimistic bias in small samples (n = 122). The full-feature LOO (54.9%) underperforming the baseline contrasts sharply with the sentiment-only LOO (60.7%), fortifying the "mix overfitting" judgment over data snooping.

11.2 Research Limitations

First, single case and external validity. Focusing solely on CATL dictates that extending these conclusions to other NEV stocks or the broad market requires caution. Modifying the weights or members of the manually selected peer portfolio could sway spillover estimates. Second, sentiment-measurement error. Neither SnowNLP nor FinBERT can fully immunize themselves against irony, clickbait, and announcement noise. Monthly averages in sparse-post months suffer volatile estimations. Third, sample-size constraints. The valid daily trading-days sample is merely 122 days. LOO accuracy confidence intervals run wide, urging careful interpretation of 10% marginal significances. Fourth, identification strategy. Linear OLS dominates correlations and mediation analyses without controlling for macro interest rates, lithium-price indices, or policy dummies. Therefore, conclusions firmly sit as "associations and predictions" rather than rigid causalities. Fifth, limited economic significance. A daily LOO peak of 60.7% (merely +5% over random baseline) is impotent for tradable strategies, and monthly explanatory power basically flattens at zero.

12 Conclusion

Taking CATL as a case study, anchored on a 2023–2026 all-month Guba footprint (28,929 posts over 42 months) and 462 sell-side reports, this paper systematically juxtaposes the impact of institutional–retail dual-channel sentiment, supply-chain competitor spillovers, and daily sentiment structures on stock returns. The primary conclusions are: first, at the monthly frequency, sentiment alone struggles to explain the leader's monthly flux. Pearson correlations for institutional and retail sentiments against concurrent and lagged returns are insignificant ($|r| < 0.16$, $p > 0.35$), rejecting the naive assumption that "forum/report averages dominate monthly trends." Second, sector price linkage is potent, yet sentiment is an unreliable mediator. Competitor equal-weight returns exhibit an $r = 0.81$ correlation and $c = 0.84$ total effect with CATL. However, Sobel tests for all four sentiment-mediation paths fail to achieve significance ($p > 0.23$), confirming that supply-chain resonance is driven by price covariance rather than the "peer price → sentiment → CATL pricing" conduit. Third, daily-frequency persistent structures harbor weak incremental

information. The correlation for consecutive high-sentiment days is marginally significant ($r = 0.196$, $p = 0.054$), with the 3–5-day group edging out the 0-day group by ~1.65%. In LOO setups, sentiment-only accuracy (60.7%) eclipses price-volume alone (59.8%) and full-mix feature arrays (54.9%).

Theoretical enlightenment. For the same target, professional institutional expressions and retail-forum noise both fail as linear pricing factors on a monthly scale. Identifying sentiment effects necessitates diving down to daily frequencies and mapping persistent structures rather than raw single-day pulses. Supply-chain analyses should position competitor returns as core explanatory variables while reserving sentiment as an auxiliary mechanism, not a default mediator.

Practical enlightenment. When tracking CATL's short-term trajectories, the weight of competitor-sector returns and lagged price-volume features should overpower daily Guba sentiment. Only when discussion heat blazes persistently high for 3–5 days does it merit inclusion in risk monitoring, yet it remains unsuitable as a standalone trading signal.

References

[1] National Development and Reform Commission (NDRC). Vigorously develop green and low-carbon industries, and promote energy conservation and carbon reduction. 2026-01-21.

[2] Department of Commerce of Heilongjiang Province, et al. Implementation rules for the car trade-in subsidy program in Heilongjiang Province. 2026-01-23. https://www.hlj.gov.cn/hlj/sjzc/202601/c00_31908235.shtml

[3] Ministry of Industry and Information Technology (MIIT). Lithium-ion battery industry specification conditions, Notice No. 14 of 2024. 2024-06-19. https://www.eesia.cn/contents/113/6922.html

[4] JunHe LLP. Analysis of the "Big and Beautiful Law": the impact of new clean-energy incentive measures on Chinese enterprises. 2025-07-07. https://www.lexology.com/library/detail.aspx?g=0ac9bcba-884c-4a17-ba4b-51a4f2f4765e

[5] Trade and Industry Department (TID). Business information bulletin No. 535/2023 — EU regulations on batteries and waste batteries. 2023-08-01. https://www.tid.gov.hk/sc/tradecircular/2023/ci5352023.html

[6] China News Service. The new energy vehicle market is still in a stage of rapid development. 2025-05-14. https://www.chinanews.com.cn/cj/2025/05-14/10415468.shtml

[7] China News Service. Two consecutive global firsts: China's automotive industry has made a crucial leap. 2026-01-18. https://www.chinanews.com.cn/cj/2026/01-18/10554032.shtml

[8] Accounting Cube. The secret behind CATL's soaring profits: how the battery giant copes with raw-material price fluctuations and market competition. 2026-01-08.

[9] Cfi.cn. Ningde Times (300750): new foreign exchange hedging quota for 2025. 2025-12-05. http://vipwt1.cfi.net.cn/newspage.aspx

[10] Car Centennial Institute. New species, new concepts, new trends: insights and forecasts on new energy vehicle consumption (2024–2025). 2025-04-06. https://www.ev100online.com/research/detail/2192

[11] China Energy News. The era of full electrification of public-transportation vehicles is upon us. 2023-02-13. https://paper.people.com.cn/zgnyb/html/2023-02/13/content_25966035.htm

[12] Sohu. CATL leads the global battery market: R&D investment and talent advantages have forged its leadership position. 2025-10-27. https://www.sohu.com/a/948024088_122145333

[13] PCauto. Will there be any problems when plug-in hybrid vehicles are used in cold northern regions? 2026-01-23. https://www.pcauto.com.cn/ask/4836586.html

[14] East Money. CATL: 2023 annual report. 2024-03-16. https://data.eastmoney.com/notices/detail/300750/AN202403151626860510.html

[15] Jinling News. Analysis of the production of CATL's Qilin battery in mid-November 2025: technical advantages and market prospects. 2025-11-11. https://www.gilin.com.cn/essence1110947.html

[16] National Energy Administration. China hydrogen energy development report. 2025-04-27. https://www.nea.gov.cn/

[17] SNE Research. Global EV battery usage for 2025. 2026-02-04. https://www.electrive.com/2026/02/04/sne-research-catl-continues-to-dominate-global-battery-market/

[18] CnEVPost. Global EV battery market share in 2025: CATL remains No. 1, BYD follows. 2026-02-04. https://cnevpost.com/2026/02/04/global-ev-battery-market-share-2025/

[19] Mordor Intelligence. China electric vehicle battery manufacturing market size and share analysis: growth trends and forecast (2026–2031). 2026. https://www.mordorintelligence.com/industry-reports/china-electric-vehicle-battery-manufacturing-market

[20] Gasgoo. CATL's offense and defense strategy. 2026-01-20. https://autonews.gasgoo.com/articles/news/catls-offense-and-defense-strategy-2007811950841176065

[21] EqualOcean. In-depth study on CATL's closed-loop recycling system. 2026-01-29. https://www.iyiou.com/data/202601291120922

[22] Soochow Securities. CATL (300750) in-depth report: leading in all aspects of NCM and LFP batteries. 2024-08. https://pdf.dfcfw.com/pdf/H3_AP202408061639151491_1.pdf

[23] Veken. Analysis of CATL's financial report: market maneuvering behind the massive cash flow. 2026-02-02. https://mp.ofweek.com/ee/a556714710667

[24] All-About-Industries. China 2026: sodium-ion batteries should achieve their breakthrough. 2026-01-15. https://www.all-about-industries.com/china-2026-sodium-ion-batteries-should-achieve-their-breakthrough-a-4fb6383e3ecc6b7ef7eafd49d515a6d7/

[25] SMM News. CATL achieves new milestone in solid-state battery R&D: team expanded to 1,000, 20 Ah sample enters trial production. 2025-11-10. https://news.metal.com/newscontent/103029009/

[26] ChinaEVHome. CATL 2025 mid-year report: total revenue reaches $25.04 billion. 2025-07-31. https://chinaevhome.com/2025/07/31/catl-2025-mid-year-report-total-revenue-reaches-25-04-billion-up-7-27-yoy/

[27] Charged EVs. CATL to deploy sodium-ion EV batteries at commercial scale in 2026. 2026-01-05. https://chargedevs.com/newswire/catl-to-deploy-sodium-ion-ev-batteries-at-commercial-scale-in-2026/

[28] Limin, A. CATL net profit jumps 41% on surging EV demand. Caixin Global, 2025-10-21. https://www.caixinglobal.com/2025-10-21/catl-q3-net-profit-jumps-41-on-surging-ev-demand-102374114.html

[29] CATL. 2024 environmental, social and governance (ESG) report. 2025, p. 85. https://www.catl.com/en/uploads/1/file/public/202505/20250514174222_ndwyqrs061.pdf

[30] OFweek. Making 200 million yuan per day, yet CATL still cannot "rest easy." 2025-11-05. https://libattery.ofweek.com/2025-11/ART-36001-8500-30673059.html

[31] The New York Times. A Hungarian town seethes over a giant Chinese battery plant. 2023-03-16. https://cn.nytimes.com/world/20230316/hungary-china-battery-plant/dual/

[32] 36Kr. CATL's new strategy in 2026: sodium-ion batteries on one hand, solid-state batteries on the other. 2026-01-19. https://eu.36kr.com/en/p/3645721778524036

[33] Securities Times. Latest assessment of the energy-storage industry: shipment growth rate of about 40% in 2026. 2025-12-10. https://stcn.com/article/detail/3532576.html

[34] ORF Online. CATL in the crossfire: how US rules are rewriting EV supply chains. 2025-10-04. https://www.orfonline.org/expert-speak/catl-in-the-crossfire-how-us-rules-are-rewriting-ev-supply-chains

[35] IDTechEx. Solid-state and polymer batteries 2025–2035: technology, forecasts, players. 2025-01-08. https://www.idtechex.com/en/research-report/solid-state-and-polymer-batteries-2025-2035-technology-forecasts-players/1033

[36] Baker, M., & Wurgler, J. Investor sentiment in the stock market. Journal of Economic Perspectives, 2007, 21(2): 129–151.

[37] Lee, C. M. C., Shleifer, A., & Thaler, R. H. Investor sentiment and the closed-end fund puzzle. The Journal of Finance, 1991, 46(1): 75–109.

[38] Tetlock, P. C. Giving content to investor sentiment: the role of media in the stock market. The Journal of Finance, 2007, 62(3): 1139–1168.

[39] Loughran, T., & McDonald, B. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. The Journal of Finance, 2011, 66(1): 35–65.

[40] Jegadeesh, N., Kim, J., Krische, S. D., et al. Analyzing the analysts: when do recommendations add value? The Journal of Finance, 2004, 59(3): 1083–1124.

[41] Barber, B. M., & Odean, T. All that glitters: the effect of attention and news on the buying behavior of individual and institutional investors. The Review of Financial Studies, 2008, 21(2): 785–818.

[42] Da, Z., Engelberg, J., & Gao, P. In search of attention. The Journal of Finance, 2011, 66(5): 1461–1499.

[43] Bae, [initials]. [et al.]. [Title to be completed — cited in text but not listed in the original draft's reference list.] 2012.

[44] Aobdia, D., Caskey, J., & Ozel, N. B. Inter-industry network structure and the cross-predictability of earnings and stock returns. Review of Accounting Studies, 2014, 19(3): 1191–1224.

[45] Baron, R. M., & Kenny, D. A. The moderator–mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. Journal of Personality and Social Psychology, 1986, 51(6): 1173–1182.

[46] Gu, S., Kelly, B., & Xiu, D. Empirical asset pricing via machine learning. The Review of Financial Studies, 2020, 33(5): 2223–2273.